

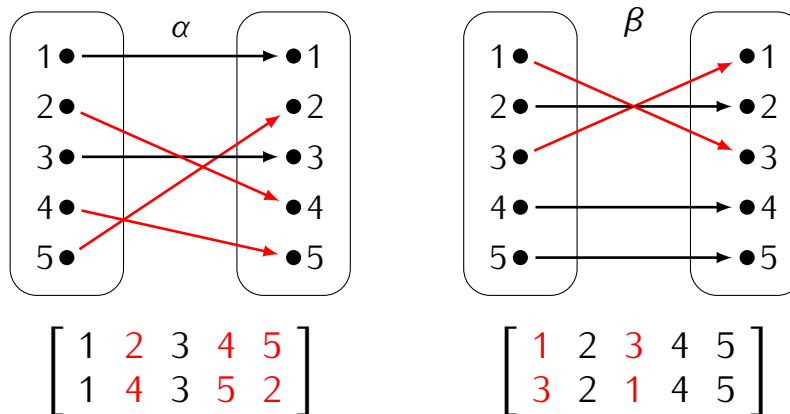
Definition 11.1

Let $\alpha \in S_n$ and let $i \in \{1, \dots, n\}$. We say that α *moves* i if $\alpha(i) \neq i$. If $\alpha(i) = i$ we say that α *fixes* i .

Definition 11.2

We will say that permutations $\alpha, \beta \in S_n$ are *disjoint* if there is no $i \in \{1, \dots, n\}$ which is moved by both α and β .

Example.



Theorem 11.3

If $\alpha, \beta \in S_n$ are disjoint permutations then $\alpha \circ \beta = \beta \circ \alpha$. Moreover,

$$\alpha \circ \beta(i) = \begin{cases} \alpha(i) & \text{if } i \text{ is moved by } \alpha \\ \beta(i) & \text{if } i \text{ is moved by } \beta \\ i & \text{otherwise} \end{cases}$$

Definition 11.4

A permutation $\alpha \in S_n$ is a *cycle of length r* (or *r -cycle*) if there are distinct elements $i_1, i_2, \dots, i_r \in \{1, 2, \dots, n\}$ such that

$$\alpha(i_1) = i_2, \quad \alpha(i_2) = i_3, \quad \dots \quad \alpha(i_{r-1}) = i_r, \quad \alpha(i_r) = i_1$$

and α fixes all other elements of $\{1, \dots, n\}$.

Theorem 11.5

Every permutation in S_n is either a cycle or a product of disjoint cycles.

Lemma 11.6

Let $\alpha \in S_n$, and let $i_0 \in \{1, \dots, n\}$ be an element moved by α . Then:

- 1) There exists $r > 1$ such that $\alpha^r(i_0) = i_0$
- 2) If $r > 1$ is the smallest integer satisfying $\alpha^r(i_0) = i_0$ then all elements

$$i_0, \alpha(i_0), \alpha^2(i_0), \dots, \alpha^{r-1}(i_0)$$

are distinct.

Poof of Theorem 11.5.

Recall that the least common multiple of integers $n_1, n_2, \dots, n_k \geq 1$ is the smallest positive integer $\text{lcm}(n_1, \dots, n_k)$ which is divisible by each of these integers.

Theorem 11.7

Assume that a permutation $\alpha \in S_n$ has a decomposition into disjoint cycles

$$\alpha = \gamma_1 \circ \cdots \circ \gamma_m$$

where γ_i is a cycle of length $r_i > 1$. Then the order of α is given by

$$|\alpha| = \text{lcm}(r_1, r_2, \dots, r_m)$$

Exercise. Compute the order of the following permutation in S_8 :

$$\alpha = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 8 & 6 & 2 & 1 & 5 & 7 & 4 \end{bmatrix}$$

Exercise. Find all possible orders of elements of S_5 .

Exercise. Compute the number of permutations of order 10 in S_8 .

Exercise. Compute the number of permutations of order 3 in S_7 .